sEMG Muscle Activation Report

Participant P3 · Gesture_1 · April 17, 2026

Overall Activation	High
Temporal Profile	Ramp-up
Consistency	Consistent
Dominant Sensors	S4
Weakest Sensors	S6
Assessment Confidence	0.95

Clinical Summary: Sustained isometric contraction with clear flexor dominance. S3 and S4 demonstrate high recruitment intensity. S8 shows a late-phase amplitude reduction, potentially indicating localized fatigue.

Key Findings:
- High flexor-extensor imbalance (2.23x)
- S3/S4 very high activation levels
- S8 amplitude decline in late phase

Session Overview

Session Duration	1202.5 s (20.0 min)
Sensors	8 (S1–S4 flexor, S5–S8 extensor)
Sampling Rate	1259.3 Hz
Session Mean RMS	0.006665 mV
Feature Source	raw_signal

Per-Sensor Activation Analysis

Muscle activity was recorded across 8 sensors. Sensors S1–S4 are positioned on the palmar/flexor compartment and S5–S8 on the dorsal/extensor compartment. RMS (Root Mean Square) reflects overall activation intensity; MNF and MDF (Mean and Median Frequency) characterise the spectral content of the EMG signal; WAMP (Willison Amplitude) indicates motor-unit firing activity.

Sensor	Mean RMS (mV)	CV	MNF (Hz)	MDF (Hz)	WAMP	Level	Profile
S1	0.003516	0.142	109.8	79.3	5	Moderate	Sustained
S2	0.004810	0.388	109.2	86.5	74	High	Sustained
S3	0.012931	0.181	143.3	122.8	2944	Very High	Ramp-up
S4	0.015383	0.139	102.9	85.0	2206	Very High	Ramp-up
S5	0.005326	0.239	126.7	103.5	274	High	Sustained
S6	0.002719	0.127	111.3	74.4	5	Moderate	Sustained
S7	0.003254	0.170	90.8	51.0	7	Moderate	Ramp-up
S8	0.005384	0.246	79.7	56.3	40	High	Ramp-down

Flexor sensors (S1–S4) shown with blue labels; extensor sensors (S5–S8) with pink.

Notable Findings per Sensor

S2: Moderate consistency

S3: High recruitment

S4: Primary driver

S5: Moderate consistency

S8: Late phase amplitude drop

Spatial Activation Patterns

Cross-channel metrics characterise how muscle activation is distributed across the forearm during the recorded gesture. A **high concentration index** indicates focal activation (one or two dominant sensors); a **low value** suggests diffuse activation. The **flexor/extensor balance** reveals which compartment drives the gesture. **Pattern stability** near 1.0 indicates the spatial pattern is maintained throughout the session, with no substantial load redistribution.

Metric	Value	Interpretation
Dominant Sensor	S4	Primary contributor
Weakest Sensor	S6	Lowest activity
Dominant-to-Weakest Ratio	5.66×	Activation disparity
Concentration Index	0.185	Focal (near 1) vs Diffuse (near 0.125)
Pattern Stability	0.994	Spatial consistency (1 = stable)

Inter-Channel Correlation	0.095	Crosstalk check (>0.8 = concern)
Co-Contraction Ratio	0.461	Agonist/antagonist balance
Flexor/Extensor Balance	2.23	>1 = flexor-dominant
Overall Temporal Profile	Ramp-up	Session-wide activation trend

Co-contraction: Not detected — Reciprocal activation pattern typical of efficient single-direction movement.

Spatial Compensation: Not detected — Stable spatial activation pattern throughout the session.

Gesture-Pattern Interpretation: Flexor group (S1-S4) shows significantly higher activation (Very High) compared to extensor group (S5-S8), consistent with palmar flexion dominance.

Motor-Unit Recruitment

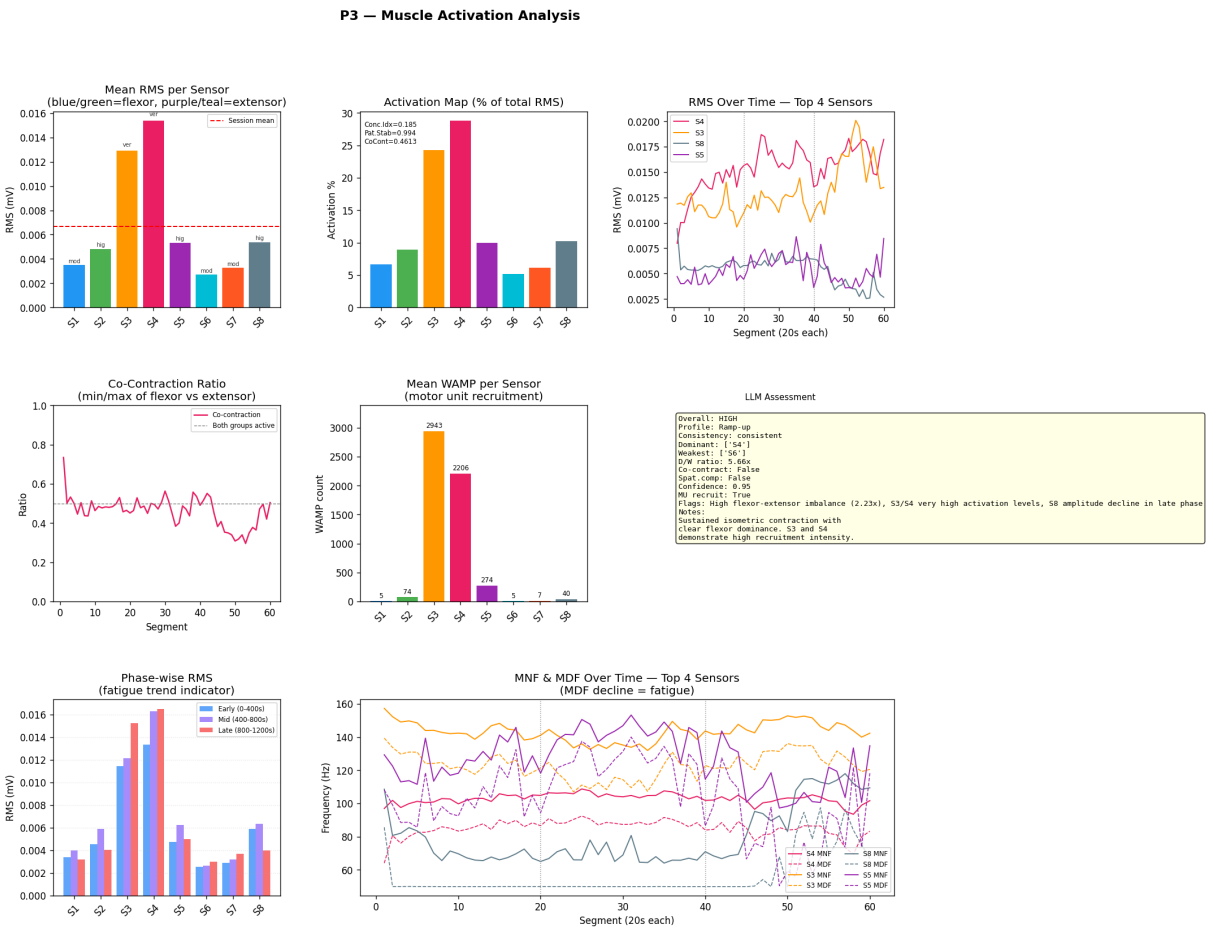
Motor-unit recruitment is inferred from the Willison Amplitude (WAMP) and zero-crossing rate. Elevated WAMP indicates frequent changes in signal amplitude, consistent with the firing of additional (often higher-threshold) motor units as effort or fatigue increases.

Recruitment Detected	Yes
Sensors Affected	S3, S4

Evidence: High WAMP values (S3: 2943.8, S4: 2206.2) indicate significant motor unit recruitment in flexor group.

Visualisation

The figure below consolidates the session analysis: per-sensor RMS amplitudes, spatial activation map, RMS time series for dominant sensors, co-contraction ratio, and frequency-content (MNF/MDF) trends over the 20-minute session.



Methodology & Limitations

Signal Processing

Raw surface EMG signals from 8 Delsys Avanti sensors were band-pass filtered (20–450 Hz, Butterworth 4th order) to attenuate motion artefacts and high-frequency noise. Twelve features were extracted per sensor in consecutive 20-second windows: RMS, MAV, IEMG, Waveform Length, Zero Crossings, Slope Sign Changes, Willison Amplitude, Peak Amplitude, Variance, Mean Frequency, Median Frequency, and Total Power. Cross-channel metrics (activation map, concentration index, pattern stability, co-contraction ratio, flexor/extensor balance) were computed from the per-sensor RMS time series.

Clinical Interpretation

Feature summaries were passed to a large language model (gemini-3.1-flash-lite-preview) along with a structured set of clinical rules (activation level thresholds, gesture patterns, co-contraction criteria, motor-unit recruitment indicators) and retrieval-augmented research context drawn from a knowledge base of sEMG literature (Konrad 2005, Phinyomark 2012, Vigotsky 2018, Nazmi 2016, Yang 2024 EMGBench).

Limitations

• **No %MVC normalisation:** Amplitudes are absolute (mV); direct cross-subject comparisons of amplitude are unreliable without a reference contraction.
• **Sustained-contraction data:** Because the gesture is held continuously, onset and duty-cycle metrics are not reported.
• **No ground-truth validation:** Findings are descriptive. Clinical decisions should consider additional context (force measurement, movement observation, patient history).
• **LLM output:** Provided as a summarisation aid, not a diagnostic tool. Human-in-the-loop review is required.

Report generated automatically by the sEMG Interpretation Pipeline. For clinical use, please corroborate findings with additional assessments.